

ASSIGNMENT NO. 1: Introduction to Databases and SQL Environment

Questions:

Q1) Write SQL commands to create a database named SchoolDB.

The program should:

1. Create a new database named SchoolDB.
2. Select the database for use.
3. Display confirmation by creating a simple table inside it.

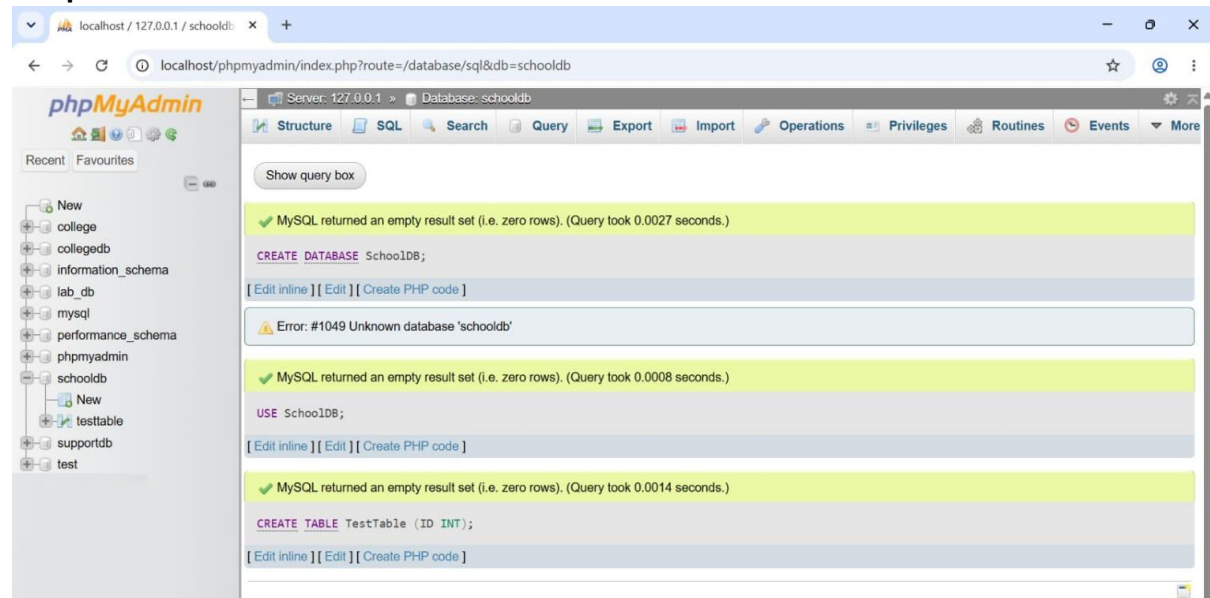
Query-

CREATE DATABASE SchoolDB;

USE SchoolDB;

CREATE TABLE TestTable (ID INT);

Output-



Q2) Create a table named Student with the following columns:

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
Age	INT	
ClassName	VARCHAR(20)	
City	VARCHAR(30)	

The program should:

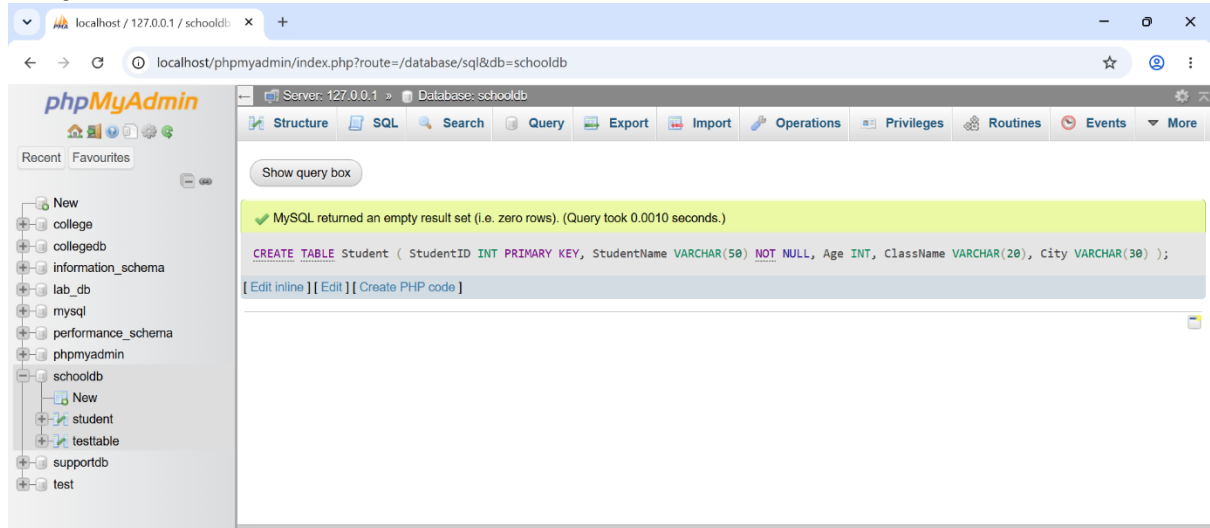
1. Create the table.
2. Define proper data types.
3. Apply primary key on StudentID.

Query-

CREATE TABLE Student (
StudentID INT PRIMARY KEY,

```
StudentName VARCHAR(50) NOT NULL,  
Age INT,  
ClassName VARCHAR(20),  
City VARCHAR(30)  
);
```

Output-



Q3) Insert at least 10 records into the Student table.

The records should include:

1. Student ID
2. Student name
3. Age
4. Class name
5. City

After inserting data, display all records using the SELECT command.

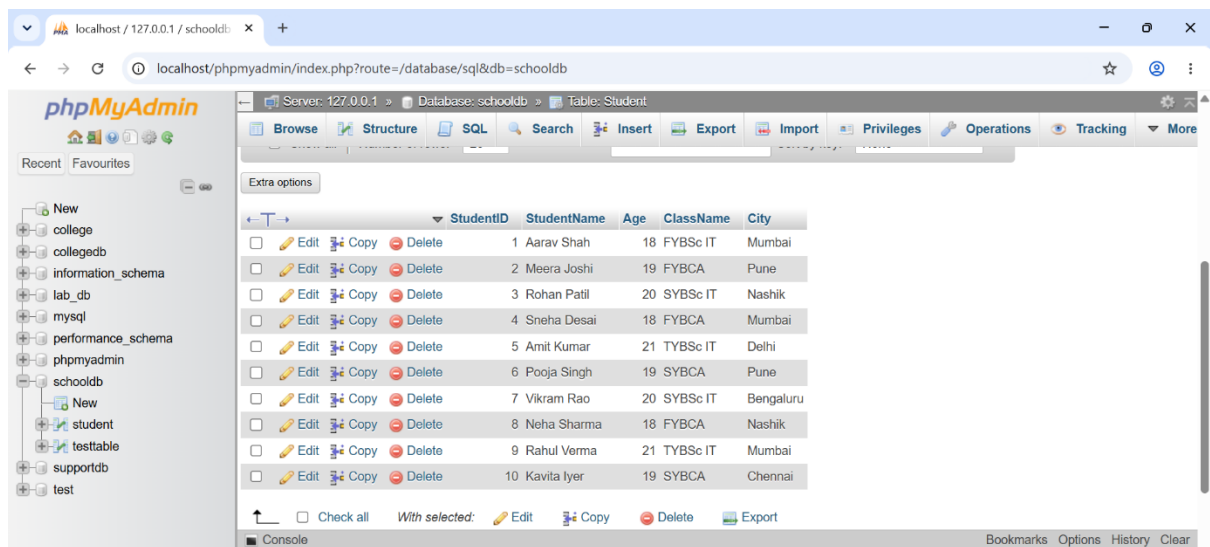
Query-

```
INSERT INTO Student (StudentID, StudentName, Age, ClassName, City)  
VALUES
```

- ```
(1, 'Aarav Shah', 18, 'FYBsc IT' , 'Mumbai'),
(2, 'Meera Joshi', 19, 'FYBCA' , 'Pune'),
(3, 'Rohan Patil', 20, 'SYBsc IT' , 'Nashik'),
(4, 'Sneha Desai', 18, 'FYBCA' , 'Mumbai'),
(5, 'Amit Kumar', 21, 'TYBsc IT' , 'Delhi'),
(6, 'Pooja Singh', 19, 'SYBCA' , 'Pune'),
(7, 'Vikram Rao', 20, 'SYBsc IT' , 'Bengaluru'),
(8, 'Neha Sharma', 18, 'FYBCA' , 'Nashik'),
(9, 'Rahul Verma', 21, 'TYBsc IT' , 'Mumbai'),
(10, 'Kavita Iyer', 19, 'SYBCA' , 'Chennai');
```

```
SELECT * FROM Student;
```

### Output-



Q4) Create a table named Course with the following fields:

| Column Name | Data Type     |
|-------------|---------------|
| CourseID    | INT           |
| CourseName  | VARCHAR(50)   |
| Duration    | VARCHAR(20)   |
| Fees        | DECIMAL(10,2) |

The program should:

1. Create the table.
2. Add a primary key on CourseID.
3. Insert at least 5 course records.
4. Display all course records.

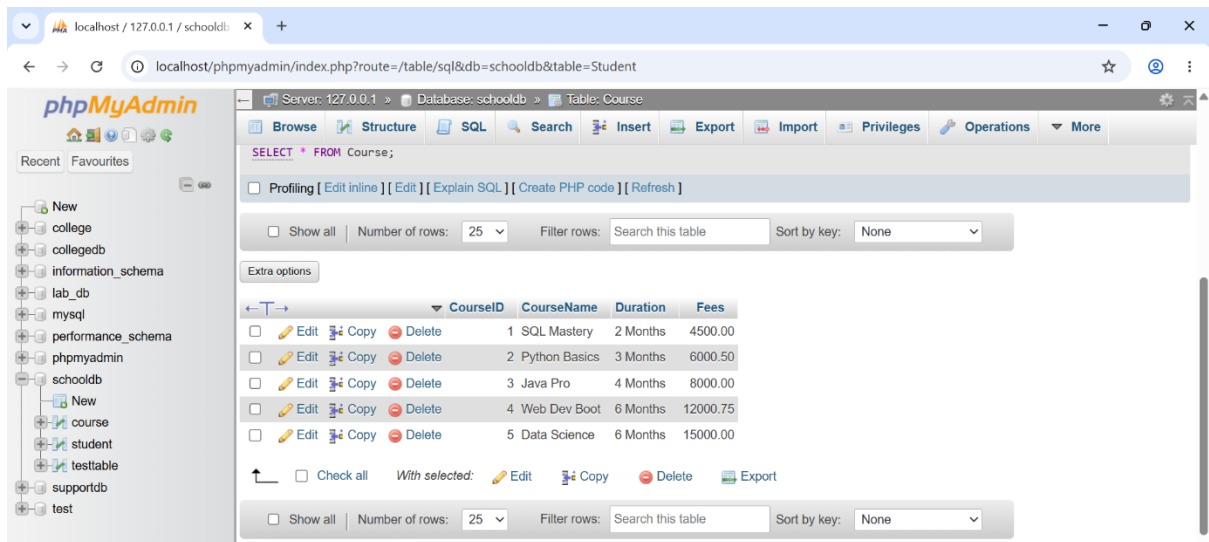
### Query-

```
CREATE TABLE Course (
 CourseID INT PRIMARY KEY,
 CourseName VARCHAR(50),
 Duration VARCHAR(20),
 FEES DECIMAL(10,2)
);
```

```
INSERT INTO Course (CourseID, CourseName, Duration, Fees) VALUES
(1, 'SQL Mastery', '2 Months', 4500.00),
(2, 'Python Basics', '3 Months', 6000.50),
(3, 'Java Pro', '4 Months', 8000.00),
(4, 'Web Dev Boot', '6 Months', 12000.75),
(5, 'Data Science', '6 Months', 15000.00);
```

```
SELECT * FROM Course;
```

### Output-



Q5) Write SQL queries to display only selected columns from the Student table.

Display:

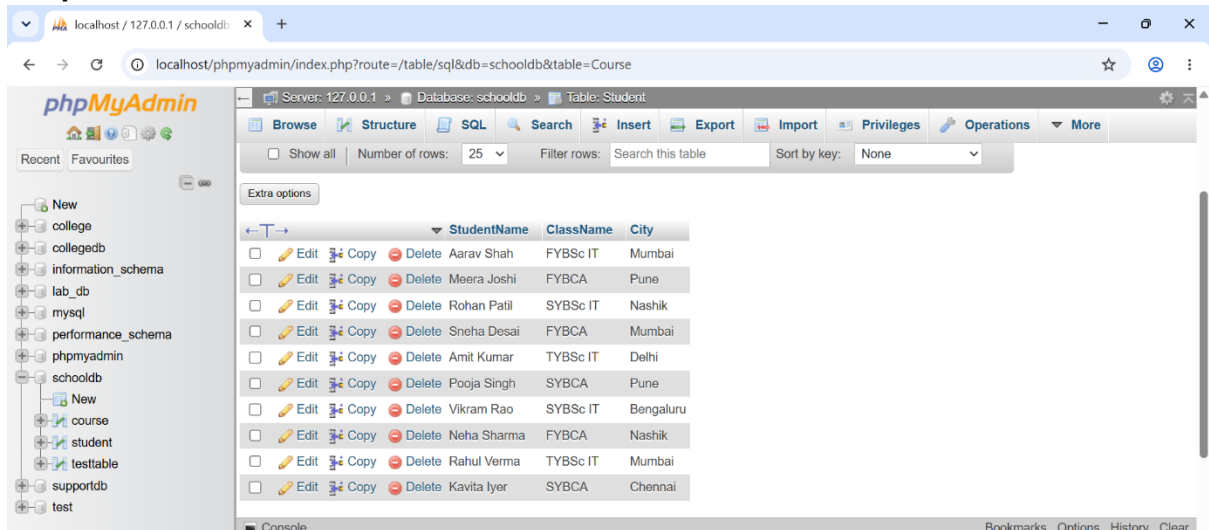
1. StudentName
2. ClassName
3. City

Expected output should show only these three columns.

**Query-**

SELECT StudentName, ClassName, City  
FROM Student;

**Output-**



Q6) Write SQL commands to view the structure of the Student table.

Use the suitable command according to the database environment:

| Database   | Command       |
|------------|---------------|
| MySQL      | DESC Student; |
| PostgreSQL | tudent        |

SQLite

PRAGMA table\_info(Student);

### Query-

-- MySQL Environment

DESC Student;

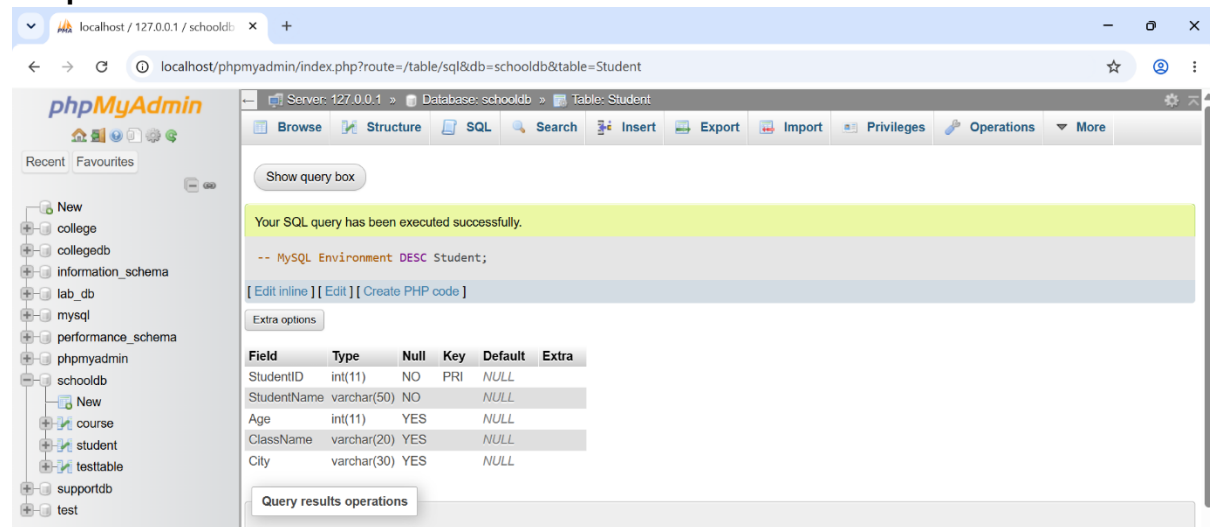
-- PostgreSQL Environment

\d Student

-- SQLite Environment

PRAGMA table\_info(Student);

### Output-



Q7) Prepare a CSV file named students.csv and import it into the Student table.

The CSV file should contain:

| StudentID | StudentName | Age | ClassName | City   |
|-----------|-------------|-----|-----------|--------|
| 1         | Aarav Shah  | 18  | FYBSc IT  | Mumbai |
| 2         | Meera Joshi | 19  | FYBCA     | Pune   |
| 3         | Rohan Patil | 20  | SYBSc IT  | Nashik |

The task should include:

1. Creating a matching table.
2. Importing the CSV file.
3. Displaying imported records using SELECT.

### Query-

LOAD DATA INFILE 'Students.csv'

INTO TABLE Student

FIELDS TERMINATED BY ','

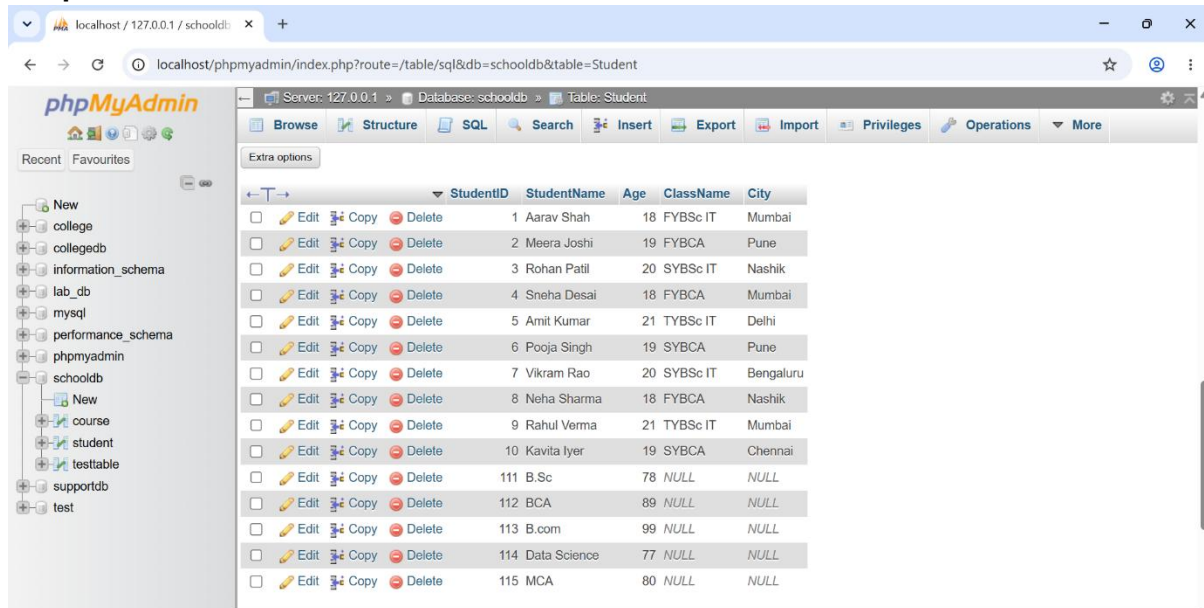
ENCLOSED BY '"'

LINES TERMINATED BY '\n'

IGNORE 1 ROWS;

SELECT \* FROM Student;

### Output-



| StudentID | StudentName  | Age | ClassName | City      |
|-----------|--------------|-----|-----------|-----------|
| 1         | Aarav Shah   | 18  | FYBSc IT  | Mumbai    |
| 2         | Meera Joshi  | 19  | FYBCA     | Pune      |
| 3         | Rohan Patil  | 20  | SYBSc IT  | Nashik    |
| 4         | Sneha Desai  | 18  | FYBCA     | Mumbai    |
| 5         | Amit Kumar   | 21  | TYBSc IT  | Delhi     |
| 6         | Pooja Singh  | 19  | SYBCA     | Pune      |
| 7         | Vikram Rao   | 20  | SYBSc IT  | Bengaluru |
| 8         | Neha Sharma  | 18  | FYBCA     | Nashik    |
| 9         | Rahul Verma  | 21  | TYBSc IT  | Mumbai    |
| 10        | Kavita Iyer  | 19  | SYBCA     | Chennai   |
| 111       | B.Sc         | 78  | NULL      | NULL      |
| 112       | BCA          | 89  | NULL      | NULL      |
| 113       | B.com        | 99  | NULL      | NULL      |
| 114       | Data Science | 77  | NULL      | NULL      |
| 115       | MCA          | 80  | NULL      | NULL      |

Q8) Create a database named EmployeeDB and a table named Employee.

The table should contain:

| Column Name  | Data Type     |
|--------------|---------------|
| EmployeeID   | INT           |
| EmployeeName | VARCHAR(50)   |
| Department   | VARCHAR(30)   |
| Salary       | DECIMAL(10,2) |
| City         | VARCHAR(30)   |

The program should:

1. Create the database.
2. Create the Employee table.
3. Insert 8 employee records.
4. Display all employee records.

### Query-

CREATE DATABASE EmployeeDB;

USE EmployeeDB;

```
CREATE TABLE Employee (
 EmployeeID INT PRIMARY KEY, EmployeeName VARCHAR(50),
 Department VARCHAR(30), Salary DECIMAL(18,2), City VARCHAR(30)
);
```

```
INSERT INTO Employee VALUES
(101, 'John Doe', 'IT', 75000.00, 'New York'),
(102, 'Jane Smith', 'HR', 65000.00, 'Chicago'),
(103, 'Sam Brown', 'Finance', 82000.00, 'Boston'),
```

```
(104, 'Lisa Ray', 'IT', 78000.00, 'Seattle'),
(105, 'Mark Lee', 'Sales', 60000.00, 'Austin'),
(106, 'Anna Kim', 'Marketing', 68000.00, 'Denver'),
(107, 'Paul Jay', 'Finance', 81880.00, 'Miami'),
(108, 'Emma Watson', 'HR', 66000.00, 'Dallas');
```

```
SELECT * FROM Employee;
```

## Output-

The screenshot shows the phpMyAdmin interface for a database named 'schooldb'. The 'Employee' table is selected, and its data is displayed in a table format. The table has columns: EmployeeID, EmployeeName, Department, Salary, and City. The data rows are as follows:

| EmployeeID | EmployeeName | Department | Salary   | City     |
|------------|--------------|------------|----------|----------|
| 101        | John Doe     | IT         | 75000.00 | New York |
| 102        | Jane Smith   | HR         | 65000.00 | Chicago  |
| 103        | Sam Brown    | Finance    | 82000.00 | Boston   |
| 104        | Lisa Ray     | IT         | 78000.00 | Seattle  |
| 105        | Mark Lee     | Sales      | 60000.00 | Austin   |
| 106        | Anna Kim     | Marketing  | 68000.00 | Denver   |
| 107        | Paul Jay     | Finance    | 81880.00 | Miami    |
| 108        | Emma Watson  | HR         | 66000.00 | Dallas   |

Q9) Write SQL queries to compare the difference between SELECT \* and selecting specific columns.

The task should include:

1. Query using SELECT \* FROM Student;
2. Query using SELECT StudentName, City FROM Student;
3. Write two lines explaining the difference between both outputs.

## Query-

--query 1: the wildcard approach

Select \*

FROM Student

--query2: the specific approach

Select StudentName, City

From student;



## Output-

Showing rows 0 - 14 (15 total, Query took 0.0010 seconds.)

-- Query 1: The Wildcard Approach `SELECT * FROM Student;`

[ Edit inline ] [ Edit ] [ Create PHP code ]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

|                          | StudentID | StudentName  | Age | ClassName | City      |
|--------------------------|-----------|--------------|-----|-----------|-----------|
| <input type="checkbox"/> | 1         | Aarav Shah   | 18  | FYBSc IT  | Mumbai    |
| <input type="checkbox"/> | 2         | Meera Joshi  | 19  | FYBCA     | Pune      |
| <input type="checkbox"/> | 3         | Rohan Patil  | 20  | SYBSc IT  | Nashik    |
| <input type="checkbox"/> | 4         | Sneha Desai  | 18  | FYBCA     | Mumbai    |
| <input type="checkbox"/> | 5         | Amit Kumar   | 21  | TYBSc IT  | Delhi     |
| <input type="checkbox"/> | 6         | Pooja Singh  | 19  | SYBCA     | Pune      |
| <input type="checkbox"/> | 7         | Vikram Rao   | 20  | SYBSc IT  | Bengaluru |
| <input type="checkbox"/> | 8         | Neha Sharma  | 18  | FYBCA     | Nashik    |
| <input type="checkbox"/> | 9         | Rahul Verma  | 21  | TYBSc IT  | Mumbai    |
| <input type="checkbox"/> | 10        | Kavita Iyer  | 19  | SYBCA     | Chennai   |
| <input type="checkbox"/> | 111       | B.Sc         | 78  | NULL      | NULL      |
| <input type="checkbox"/> | 112       | BCA          | 89  | NULL      | NULL      |
| <input type="checkbox"/> | 113       | B.com        | 99  | NULL      | NULL      |
| <input type="checkbox"/> | 114       | Data Science | 77  | NULL      | NULL      |
| <input type="checkbox"/> | 115       | MCA          | 80  | NULL      | NULL      |

Showing rows 0 - 14 (15 total, Query took 0.0009 seconds.)

-- Query 2: The Specific Approach `SELECT StudentName, City FROM Student;`

[ Edit inline ] [ Edit ] [ Create PHP code ]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

|                          | StudentName  | City      |
|--------------------------|--------------|-----------|
| <input type="checkbox"/> | Aarav Shah   | Mumbai    |
| <input type="checkbox"/> | Meera Joshi  | Pune      |
| <input type="checkbox"/> | Rohan Patil  | Nashik    |
| <input type="checkbox"/> | Sneha Desai  | Mumbai    |
| <input type="checkbox"/> | Amit Kumar   | Delhi     |
| <input type="checkbox"/> | Pooja Singh  | Pune      |
| <input type="checkbox"/> | Vikram Rao   | Bengaluru |
| <input type="checkbox"/> | Neha Sharma  | Nashik    |
| <input type="checkbox"/> | Rahul Verma  | Mumbai    |
| <input type="checkbox"/> | Kavita Iyer  | Chennai   |
| <input type="checkbox"/> | B.Sc         | NULL      |
| <input type="checkbox"/> | BCA          | NULL      |
| <input type="checkbox"/> | B.com        | NULL      |
| <input type="checkbox"/> | Data Science | NULL      |
| <input type="checkbox"/> | MCA          | NULL      |

Q10) Create a complete SQL script for a mini college database.

The script should include:

1. Creating database MiniCollegeDB.



2. Creating tables Student, Course, and Faculty.
3. Adding suitable columns and data types.
4. Inserting at least 5 records in each table.
5. Displaying all records from each table.
6. Viewing the structure of each table.

### Query-

```
CREATE DATABASE MiniCollegeDB;
USE MiniCollegeDB;
```

```
CREATE TABLE Student (SID INT PRIMARY KEY, SName VARCHAR(50));
CREATE TABLE Course (CID INT PRIMARY KEY, CName VARCHAR(50));
CREATE TABLE Faculty (FID INT PRIMARY KEY, FName VARCHAR(50));
```

```
INSERT INTO Student
VALUES (1,'Ali'), (2, 'Ben'), (3, 'Cam'), (4, 'Dan'), (5, 'Eve');
INSERT INTO Course
VALUES (10, 'Math'), (11, 'Sci'), (12, 'Art'), (13, 'Lit'), (14, 'CS');
INSERT INTO Faculty
VALUES (100, 'Dr.X'),(101, 'Dr.Y'),(102,'Dr.Z'), (103, 'Dr.A'), (104, 'Dr.B');
```

```
SELECT * FROM Student; SELECT * FROM Course; SELECT * FROM Faculty;
```

```
DESC Student; DESC Course; DESC Faculty;
```

### Output

The screenshot shows the phpMyAdmin web interface. The left sidebar displays a database structure tree with 'schooldb' selected, containing tables 'course', 'student', and 'testtable'. The main panel shows the 'Table: Faculty' view. A green status bar indicates 'Showing rows 0 - 4 (5 total, Query took 0.0010 seconds.)'. Below this, the SQL query 'SELECT \* FROM Student;' is entered. The table view shows 5 rows of data with columns 'SID' and 'SName'.

| SID | SName |
|-----|-------|
| 1   | All   |
| 2   | Ben   |
| 3   | Cam   |
| 4   | Dan   |
| 5   | Eve   |

localhost / 127.0.0.1 / schooldb: x +

localhost/phpmyadmin/index.php?route=/table/sql&db=schooldb&table=Student

phpMyAdmin

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Server: 127.0.0.1 Database: schooldb Table: Faculty

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Bookmark this SQL query

Showing rows 0 - 4 (5 total, Query took 0.0005 seconds.)

SELECT \* FROM Course;

Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

|                                                                                             | CID | CName |
|---------------------------------------------------------------------------------------------|-----|-------|
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 10  | Math  |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 11  | Sci   |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 12  | Art   |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 13  | Lit   |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 14  | CS    |

localhost / 127.0.0.1 / schooldb: x +

localhost/phpmyadmin/index.php?route=/table/sql&db=schooldb&table=Student

phpMyAdmin

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  - student
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- supportdb
- test

Server: 127.0.0.1 Database: schooldb Table: Faculty

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Bookmark this SQL query

Showing rows 0 - 4 (5 total, Query took 0.0012 seconds.)

SELECT \* FROM Faculty;

Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

|                                                                                             | FID | FName |
|---------------------------------------------------------------------------------------------|-----|-------|
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 100 | Dr.X  |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 101 | Dr.Y  |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 102 | Dr.Z  |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 103 | Dr.A  |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 104 | Dr.B  |

localhost / 127.0.0.1 / schooldb: x +

localhost/phpmyadmin/index.php?route=/table/sql&db=schooldb&table=Student

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Server: 127.0.0.1 Database: schooldb Table: Faculty

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Bookmark this SQL query

Your SQL query has been executed successfully.

DESC Student;

[ Edit inline ] [ Edit ] [ Create PHP code ]

Extra options

| Field | Type        | Null | Key | Default | Extra |
|-------|-------------|------|-----|---------|-------|
| SID   | int(11)     | NO   | PRI | NULL    |       |
| SName | varchar(50) | YES  |     | NULL    |       |

localhost / 127.0.0.1 / schooldb

localhost/phpmyadmin/index.php?route=/table/sql&db=schooldb&table=Student

phpMyAdmin

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Server: 127.0.0.1 > Database: schooldb > Table: Faculty

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Bookmark this SQL query

Your SQL query has been executed successfully.

DESC Course;

[ Edit inline ] [ Edit ] [ Create PHP code ]

Extra options

| Field | Type        | Null | Key | Default | Extra |
|-------|-------------|------|-----|---------|-------|
| CID   | int(11)     | NO   | PRI | NULL    |       |
| CName | varchar(50) | YES  |     | NULL    |       |

localhost / 127.0.0.1 / schooldb

localhost/phpmyadmin/index.php?route=/table/sql&db=schooldb&table=Student

phpMyAdmin

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- information\_schema
- lab\_db
- minicollegedb
- mysql
- performance\_schema
- phpmyadmin

Server: 127.0.0.1 > Database: schooldb > Table: Faculty

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More

Bookmark this SQL query

Your SQL query has been executed successfully.

DESC Faculty;

[ Edit inline ] [ Edit ] [ Create PHP code ]

Extra options

| Field | Type        | Null | Key | Default | Extra |
|-------|-------------|------|-----|---------|-------|
| FID   | int(11)     | NO   | PRI | NULL    |       |
| FName | varchar(50) | YES  |     | NULL    |       |